ELEC-H310 – Digital electronics
Th05: Number encodings & Boolean algebra

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v3.5

Today

1. Numbers representation
2. Base conversions
3. Arithmetic operations
4. Negative numbers
5. Boolean algebra

1. Numbers representation

Real (decimal) numbers representation

- Example of a real number, in base 10:

$$(1372.6450)_{10}$$

- Two parts, split with a decimal point (. or , are allowed):

1372	.	6450
Integer $10^3 10^2 10^1 10^0$	Decimal point	Decimal $10^{-1} 10^{-2} 10^{-3} 10^{-4}$
$1 \times 10^3 + 3 \times 10^2$ $+7 \times 10^1 + 2 \times 10^0$	$+$	$6 \times 10^{-1} + 4 \times 10^{-2}$ $+5 \times 10^{-3} + 0 \times 10^{-4}$

- This is a fixed point representation; used for fractional and/or non-integer part of of decimal numbers with predefined & fixed number of digits (bits) decided at circuit design time
  - ▹ Opposed to floating point representation, used for real numbers in most digital systems (processors/ALUs) these days; not covered here

General form (for any base)

- Number N in base r, noted N_r:

$$N_r = (a_n a_{n-1} a_{n-2} \ldots a_0 b_1 b_m)_r$$

where:

  - $a_0, \ldots, a_n$ – are $n+1$ digits used for integer part I
  - $b_1, \ldots, b_m$ – are m digits used for decimal part D
  - in the above different digits are concatenated

- Number N_r is then written as:

$$\begin{aligned} N_r &= a_n \cdot r^n + a_{n-1} \cdot r^{n-1} + a_{n-2} \cdot r^{n-2} + \ldots + a_0 \cdot r^0 \\ &\quad + b_1 \cdot r^{-1} + b_2 \cdot r^{-2} + \ldots + b_m \cdot r^{-m} \\ &= \sum_{i=0}^{i=n} a_i \cdot r^i + \sum_{j=1}^{j=m} b_j \cdot r^{-j} \\ &= I + D \end{aligned}$$

Useful bases

- In these lectures we will consider useful bases
  - Base $r = 10$ – because of us, humans, and
  - Bases $r = 2$ – binary, $r = 8$ – octal and $r = 16$ – hexadecimal
    - These 3 are used in digital systems and computers, they are power of 2 ($2^1 \rightarrow$ binary, $2^3 \rightarrow$ octal, $2^4 \rightarrow$ hexadecimal)
- Each base defines a set of symbols (digits) $\mathtt{D}$ used to describe numbers using positional notation:
  - decimal $\mathtt{D} \in \{\mathtt{0,1,2,3,4,5,6,7,8,9}\}$
  - binary $\mathtt{D} \in \{\mathtt{0,1}\}$
  - octal $\mathtt{D} \in \{\mathtt{0,1,2,3,4,5,6,7}\}$
  - hexadecimal $\mathtt{D} \in \{\mathtt{0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}\}$
- It is important to learn how to switch from one base to another – conversions, covered in the next Chapter; and how to perform basic arithmetic computations (Chapter 3 of this lecture)

General form (for any base) 2/2

- Looking at the integer part of the number N in base r:

$$N_r = \sum_{i=0}^{i=n} a_i \cdot r^i + \sum_{j=1}^{j=m} b_j \cdot r^{-j} = I + D$$

- We can define:
  - Least Significant Digit for $i = 0$
  - Most Significant Digit for $i = n$
  - For binary base, the above becomes Least Significant Bit (LSB) and Most Significant Bit (MSB)
- Knowing if the word starts with the MSB or LSB defines the **direction** in which we should read that word; in these lectures most of the time MSB is on the left and LSB is on the right (unless specified otherwise)
  - If you are not sure, ask, because $(1011)_2$ could be interpreted as 11 or 13 and this is obviously very much different

Numbers in different (useful) bases

Decimal (10)	Binary(2)	Octal (8)	Hexadecimal (16)
0	00000	0	0
1	00001	1	1
2	00010	2	2
3	00011	3	3
4	00100	4	4
5	00101	5	5
6	00110	6	6
7	00111	7	7
8	01000	10	8
9	01001	11	9
10	01010	12	A
11	01011	13	B
12	01100	14	C
13	01101	15	D
14	01110	16	E
15	01111	17	F
16	10000	20	10

2. Base conversions

Method of successive divisions – Dec2Bin, integer

- Number N (base 2), coded with 4 digits (integer part) written as:

$$
\begin{aligned}
N &= \sum_{i=0}^{i=3} a_i \cdot r^i \\
N &= 1 \cdot a_0 + 2 \cdot a_1 + 2 \cdot (2 \cdot a_2) + 2 \cdot (2 \cdot (2 \cdot a_3)) \\
N &= 2 \cdot (2 \cdot (2 \cdot (a_3) + a_2) + a_1) + a_0
\end{aligned}
$$

- Last expression can be reorganized and written as:

$$
\begin{aligned}
(N - a_0) : 2 &= 2 \cdot (2 \cdot (a_3) + a_2) + a_1 && (1) \\
((N - a_0) : 2 - a_1) : 2 &= 2 \cdot (a_3) + a_2 && (2) \\
(((N - a_0) : 2 - a_1) : 2) - a_2) : 2 &= a_3 && (3)
\end{aligned}
$$

- Eq. (1) gives us LSB, (2) next bit and (3) the last bit or MSB

Example Dec2Bin, integer part

Convert following number from base 10 to base 2: $(245)_{10} = (?)_2$

- Build a table: each line represents one division, so one result bit

Computation	Decimal	Base	Remainder	Read direction
$\vert$	245	:2	1	$\uparrow$
$\vert$	122	:2	0	$\vert$
$\vert$	61	:2	1	$\vert$
$\vert$	30	:2	0	$\vert$
$\vert$	15	:2	1	$\vert$
$\vert$	7	:2	1	$\vert$
$\vert$	3	:2	1	$\vert$
$\vert$	1	:2	1	$\vert$
$\downarrow$	0			$\vert$

- Note the **computation direction** (from top to bottom); result is read from bottom to top $\rightarrow$ **read direction**; this is because the first division gave LSB and last MSB
- Result is written as: $(245)_{10} = (11110101)_2$

Successive multiplications – Dec2Bin, real part

- Same reasoning applied for decimal part but instead of divisions we should use multiplications; example: $(0.345)_{10} = (?)_2$

Computation	Decimal	Base	Remainder	Direction
|	.345	x2	0	|
|	.690	x2	1	|
|	.380	x2	0	|
|	.760	x2	1	|
|	.520	x2	1	|
|	.040	x2	0	|
|	.080	x2	0	|
|	.160	x2	0	|
|	.320	x2	0	|
|	.640	x2	1	|
↓	.280	x2	0	↓

- When do we stop will depend on the number of digits that we want to accept; given by fixed point and number of bits reserved
- Note result reading direction, so for 10 digits: $(.0101100001)_2$

Conversion Bin2Dec, integer & decimal part

- Conversions in that direction are straightforward
- Convert the integer part $(\texttt{11110101})_2 = (?)_{10}$

$$
\begin{aligned}
&= 1\times 2^7 + 1\times 2^6 + 1\times 2^5 + 1\times 2^4 + 0\times 2^3 + 1\times 2^2 + 0\times 2^1 + 1\times 2^0 \\
&= 128 + 64 + 32 + 16 + 0 + 4 + 0 + 1 \\
&= 245
\end{aligned}
$$

- Convert the decimal part $(.\texttt{0101100001}\ldots)_2 = (?)_{10}$

$$
\begin{aligned}
&= 0\times 2^{-1} + 1\times 2^{-2} + 0\times 2^{-3} + 1\times 2^{-4} + 1\times 2^{-5} + 0\times 2^{-6} \\
&\quad +0\times 2^{-7} + 0\times 2^{-8} + \ldots \\
&= 0\times 1/2 + 1\times 1/4 + 0\times 1/8 + 1\times 1/16 + 1\times 1/32 + 0\times 1/64 + \ldots \\
&= (8+2+1)/32 = 11/32 = 0.34375
\end{aligned}
$$

Conversion from Dec2Oct (and back)

- Successive divisions/multipl. method can be applied to any base!
- Convert $(245)_{10} = (?)_8$

Computation	Decimal	Base	Remainder	Direction
$\vert$	245	:8	5	$\uparrow$
$\vert$	30	:8	6	$\vert$
$\vert$	3	:8	3	$\vert$
$\downarrow$	0			$\vert$

so, the result is $(245)_{10} = (365)_8$

- Converting back to decimal from octal is easy:

$$\begin{aligned}(365)_8 &= 5 \times 8^0 + 6 \times 8^1 + 3 \times 8^2 \\ &= 5 + 48 + 192 \\ &= (245)_{10}\end{aligned}$$

Conversion from Dec2Hex (and back)

- Perform the following conversion: $(245)_{10} = (?)_{16}$

Computation	Decimal	Base	Remainder	Lecture
$\vert$	245	:16	5	$\uparrow$
$\vert$	51	:16	F	$\vert$
$\downarrow$	0			$\vert$

so, the result is: $(\mathtt{245})_{\mathtt{10}} = (\mathtt{F5})_{\mathtt{16}}$

- Conversion back from hex to dec (another example):

$$
\begin{aligned}
(A1F.1C)_{16} &= (?)_{10} \\
&= A \times 16^2 + 1 \times 16^1 + F \times 16^0 + 1 \times 16^{-1} + C \times 16^{-2} \\
&= 10 \times 16^2 + 1 \times 16^1 + 15 \times 16^0 + 1 \times 16^{-1} + 12 \times 16^{-2} \\
&= 2560 + 16 + 15 + 28/256 \\
&= 2591.1093...
\end{aligned}
$$

- OK but cumbersome; luckily for bases that are power of 2, there is a much easier way, with (almost) no computational hustle

Conversion technique for power 2 bases

- For power 2 bases (binary, octal, hex), conversions can be done by grouping binary digits into packs of 3 bits for octal, and 4 bits for hexadecimal, starting from LSB (this is very important)
- Remember also that any number can be completed with an arbitrary number of '0"s to the left of the MSD (MSB for binary)
- Conversions – binary (left) & hex (right) to other power 2 bases:

Base 2	011110101		
3 bits bins	011	110	101
Base 8	3	6	5
4 bits bins	0000	1111	0101
Base 16	0	F	5

Base 16	1F2		
Base 2	0001	1111	0010
3 bits bins	111	110	010
Base 8	7	6	2
Base 10	498		

Example of conversion from dec to bin, oct & hex

Perform the following conversions: $(378)_{10} = (?)_2 = (?)_8 = (?)_{16}$

- First convert from dec to bin: $(378)_{10} = (101111010)_2$
  - We need 9 bits, so for oct we do not need to append any 0's to the left; for hex we need to add 3 bits set to 0 (12 bits or 3 hex digits)
- Then from bin to oct and hex:

Base 2	101111010		
3 bits bins	101	111	010
Base 8	5	7	2
4 bits bins	0001	0111	1010
Base 16	1	7	A

- Final results is then written as:
  $(378)_{10} = (101111010)_2 = (572)_8 = (17A)_{16}$

Arbitrary conversions

- How to convert a number N in base p to a number X in base q?

$$N_p \rightarrow (X)_q$$

  It can be done using the method of successive divisions for the integer part and multiplications for the decimal part; but this could be confusing & error prone; you will be doing do it during the exercises, but not for the exam $\rightarrow$ we adopt simplified way

- Simply perform intermediate conversion to base 10 and then move to other base:

$$N_p \rightarrow N_{10} \rightarrow (X)_q$$

- For useful basis start with base 2 first, the others will follow

$$N_{10} \rightarrow (?)_2 \rightarrow (?)_8 \rightarrow (?)_{10}$$

Example of arbitrary conversion

Perform conversion from base 8 to base 5: $(25.34)_8 = (?)_5$

- First convert base 8 to base 10:

$$(25.34)_8 = (?)_{10} = 2 \times 8 + 5 \times 1 + 3 \times 8^{-1} + 4 \times 8^{-2} = (21.4375)_{10}$$

- Then convert from base 10 to base 5: $(21.4375)_{10} = (?)_5$

21	:5	1	↑	→	|	.4375	x5	2.1875	2
4	:5	4	|		|	.1875	x5	0.9375	0
0			|		|	.9375	x5	4.6875	4
			|		|	.6875	x5	3.4375	3
			|		|	.4375	x5	2.1875	2
			|		↓	.1875	x5	0.9375	0

- Result: $(21.4375)_{10} \rightarrow (41.204320)_5$

3. Arithmetic operations

Binary addition of two 1-bit words

- Assume two 1-bit words a, b; addition of these two words $s = a + b$ will have 2 bits of the result $(s_1 s_0)$ because:

$$(1)_{10} + (1)_{10} = (2)_{10} = (10)_2$$

- For all combinations of a,b we can compute the sum s that is a two bit word: sum bit (s_0) & carry bit (s_1)

a	b	s_1	s_0
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

- Above specification can be translated into a logic circuit called half-adder; we will learn later how to derive the logic equations

Binary addition of three 1-bit words

- Previous circuit is extended with an extra bit to calculate:

$$s = a + b + c$$

- Result uses a 2 bit word:

$$\begin{aligned}(1)_{10} + (1)_{10} + (1)_{10} &= (3)_{10} \\ &= (11)_2\end{aligned}$$

a	b	c	s_1	s_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

- For all combinations of a, b and c we can compute the sum s represented in the table
- Note that the bit s_1 is still called carry bit
- Above specification can be translated into a logic circuit called full-adder, another important circuit

Binary addition of 2 words a, b of n-bits

- The sum s of two words of arbitrary length (n-bits) can be done using series of additions starting from the LSB
- Result, word s is encoded with $n+1$ bits ($2 \times 2^n = 2^{n+1}$)

Carry	c_{n-1}	c_{n-2}	$\dots$	c_1	c_0	
A		a_{n-1}	$\dots$	a_2	a_1	a_0
B		b_{n-1}	$\dots$	b_2	b_1	b_0
S		s_{n-1}	$\dots$	s_2	s_1	s_0

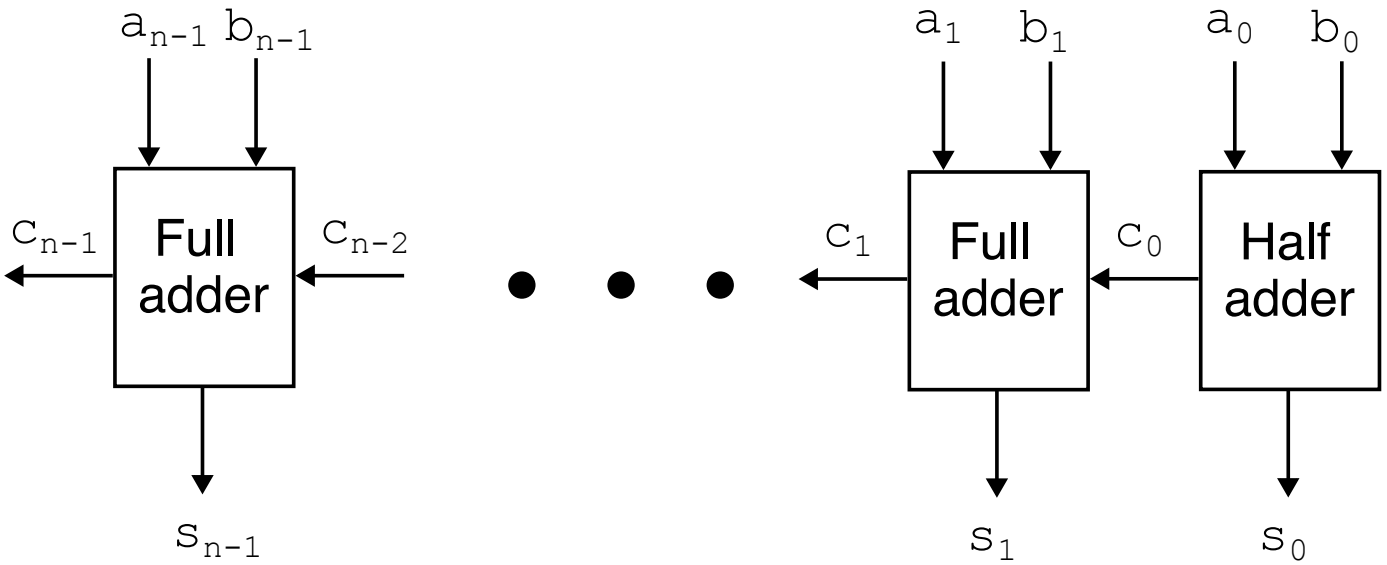
- With:

$$\begin{aligned} c_0 s_0 &= a_0 + b_0 \\ c_i s_i &= a_i + b_i + c_{i-1} \quad \forall i \in \{1, \dots, n-1\} \end{aligned}$$

Note that computation $i+1$ must wait for i to complete, due to serial nature of computation!

Addition circuit for 2 words of n bits

- LSB sum can be computed using one half-adder circuit using LSBs of a, b as operands
- Other $n-1$ bits can be computed using $n-1$ full adders $\rightarrow$ n circuits are assembled in series for a complete n-bit adder
- This circuit is known as ripple carry adder, carry needs to propagate from LSB to MSB



Above circuit will not be good for $n >>$!

Can you explain why?

Addition example with two 8-bits words

Compute $s = a + b = 236 + 170 = ?$

- We perform this operation bit-by-bit assuming two 8-bit words

Bit position (p)	8	7	6	5	4	3	2	1	0
Bit weight (2^p)	256	128	64	32	16	8	4	2	1
Carry bit	1	1	1	0	1	0	0	0	
a		1	1	1	0	1	1	0	0
b		1	0	1	0	1	0	1	0
Sum ($s_7 \ldots s_0$)	1	1	0	0	1	0	1	1	0

- Result $(406)_{10}$ can be easily verified $(406)_{10} = (1\ 1001\ 0110)_2$
- Result needs 9 bits; in systems with fixed word size (e.g., 8-bits) this will cause an overflow (bit weight 8); if the result is to be stored in an 8-bits register, memory, etc. it will be truncated
  - ▹ What will happen if this occurs in a desktop/embedded system?

4. Negative numbers

Overview

- Representation – here we cover the following forms for the representation of negative numbers:
  a. Sign and Magnitude (SM)
  b. One's complement (C1)
  c. Two's complement (C2)
- Conversions – we will look into methods that allow us to convert numbers to and from any of the above (base 10 only)
- Basic arithmetic operations – here we will discuss only C2 in details, since this is what every single computing machine is using today, the others forms are mentioned for information only

a. Signed & Magnitude (SM) representation

- Out of n bits used to represent a number is SM representation, one bit is reserved for the sign
- By convention we say:
  - 0 – for positive numbers
  - 1 – for negative numbers
  - $n-1$ bits are reserved for the magnitude
- Example of representation using one word of 8 bits:
  - We have 1 bit for the sign
  - There are 7 bits for magnitude; so 2^7 values
  - Maximum is $2^7-1=127$, minimum is -127 & therefore we can represent from `-127` to `+127` since:

```
1 1111111 (-127)      0 1111111 (+127)
```

Attention! So we have $2 \times 127 + 1 = 255$ while $2^8 = 256$

How come?

a. SM representation has 2 zeros

One '0' with $(+)$ and the other '0' with $(-)$ sign

Sign	Decimal	Binary	SVA
$+$	0	0 0000000	$(+)0$
$+$	1	0 0000001	1
$+$	...	...	
$+$	127	0 1111111	$+127$
$-$	128	1 0000000	$(-)0$
$-$	129	1 0000001	-1
$-$	...	...	
$-$	255	1 1111111	-127

a. Arithmetics using SM – subtraction

- We need to compare both signs to decide how to proceed, and we need to compare the magnitude to decide on computation direction (and the result sign)

  Example:

  - A, B are of the same $(+)$ positive sign, and
  - if $A > B$, then $A - B$, or
  - it $A < B$, then $B - A$

- SM advantages – easy to understand, we normally use this mode of representation with base 10 (negative sign before the number); very easy to convert to and from
- SM disadvantages – there are 2 different symbols with the same meaning; computations are more complex and slow since we need the adjustments that require special circuits, physical implementation (HW) is more expensive, more power, possibly lower performance

b. One's complement (C1)

- Motivation – compute subtraction like addition
- Same circuit could do two different operations (addition & subtraction) so nice optimization for HW; & works for any base
- Assume two words A & B in base $r = 10$ encoded using m digits:

$$\begin{aligned} A - B &= A + (-B) \\ &= A + (r^m - B) = A + \overline{B} \end{aligned}$$

- In the above $\overline{B}$ is the complement to base 10 (or radix complement) and $B + \overline{B} = r^m \rightarrow$ one's complement (C1)
- Practical example of computation for $r = 10, m = 3, r^m = 1000$

```
+87 --->     87              --->    87
-53 --->  1000-53=947        --->   947
-------------------------------------
                                   1034
```

b. Two's complement (C2)

- To do $A - B$ we still need to compute $\overline{B} = (r^m - B)$, so we need a subtraction operation that we wanted to get rid of
- Luckily we can re-organize B by adding and subtracting 1:

$$\overline{B} = (r^m - B) = ((r^m - 1) - B) + 1$$

where $(r^m - 1) - B$ is a complement of each digit of B noted:

$$\begin{aligned}(r^m - 1) - B &= ((r-1)(r-1)\ldots(r-1)) - (b_{m-1}b_{m-2}\ldots b_0) \\ &= ((r-1) - b_{m-1})((r-1) - b_{m-2})\ldots((r-1) - b_0) \\ &= b'_{m-1}b'_{m-2}\ldots b'_0\end{aligned}$$

  - ▹ Note that in the above, terms on the right are concatenation and not multiplication! b'_i are complement of each digit
- Complement of each digit in base 2 is simple to implement in HW: each digit is pushed through an inverter, basic logic component

b. Summary – two variants of the complement

a) One's complement (C1)

$$(r^m - 1) - B$$

Decimal	Binary	C1
0	00000000	(+)0
1	00000001	1
...	...	...
127	01111111	+127
128	10000000	−127
129	10000001	−126
...	...	...
255	11111111	(−)0

We still have 2 zeros:
00000000 &
11111111

b) Two's complement (C2)

$$(r^m - 1) - B + 1$$

Decimal	Binary	C2
0	00000000	0
1	00000001	1
...	...	...
127	01111111	+127
128	10000000	−128
129	10000001	−127
...	...	...
255	11111111	−1

There is a skew,
but only one zero!

Practical conversions to SM, C1 & C2

- Absolute value of the number is first converted to binary
  - ▹ You should use normal base 10 to 2 conversion (methods of successive divisions/multiplications)
- Note that you should complete the number with '0's to get the desired length of the word: imagine that you have to encode a word using m bits and you get n bits from the conversion, then $m-n$ '0's are placed in front of the MSB
- For SM you should keep $m-1$ bits and use bit weight m to encode the sign (0 for positive numbers, 1 for negative)
- If the number is negative, every bit of the word encoded in m bits, is inverted to get One's complement (C1); then we add 1 to C1 and derive Two's complement (C2)
- If the number is positive, just rewrite using m bits

Example of conversion to SM, C1 & C2

- Convert $(57)_{10}$ & $(-23)_{10}$ to SM, C1 & C2 using 8 bits
- For positive numbers (57_{10}) → just use binary encoding!; note completing with '0's

Absolute value	SM (8-bits)	C1 (8-bits)	C2 (8-bits)
11_1001	0011_1001	0011_1001	0011_1001

- For negative numbers (-23) → convert to binary, add sign bit for SM, invert for C1 & add 1 for C2

Absolute value	SM (8-bits)	C1 (8-bits)	C2 (8-bits)
1_0111	1_001_0111	0001_0111	
		1110_1000	1110_1001

Arithmetics with negative numbers

- We have seen that we can do arithmetics using SM & C1
  - We will not go deeper into these here $\rightarrow$ too many particular cases to analyse in order to decide on how to proceed
- However in two's complement (C2) arithmetics is straightforward!
- Imagine that you want to compute $C = A - B$
  - Positive number A is converted into C2 (just binary conversion)
  - Negative number B is converted into C2; absolute value is converted to binary, we apply bit-wise inversion, & add 1 to the C1
  - Perform subtraction as an addition
  - Check your result for the overflow with the following rule – overflow bit is ignored as long as
    - Result C is in the range: $-r^m < C < r^{m-1}$
      for 8 bits this is : $-2^7 < C < 2^7 - 1$, so: $-128 < C < 127$
    - Last two bits have the same value

Example of subtraction using C2

- Calculate using C2 the following:
  $C = A - B = 57 - 23 = ?$
- We convert and do the addition of a negative number:

	Overflow	Bits 7-4	Bits 3-0	Decimal
Carry	1	1111	001–	
A		0011	1001	$+(57)_{10}$
B		1110	1001	$-(23)_{10}$
C	1	0010	0010	$+(34)_{10}$

- Check the result:
  - Result is OK, we can ignore the overflow (see below)
  - Two last bits (MSB) of the carry are the same (here 11)
  - The result is in the range $-128 < 34 < 127$

5. Boolean algebra

Definition of Boolean algebra

- Boolean algebra is a quadruplet $\{B, ', \cdot, +\}$ where:
  - $B \rightarrow$ is a two-element set
  - $'$, $\cdot$ & $+$ are respectively $\rightarrow$ complement[1], `AND` and `OR` operators
- Depending on how we define the set B, and the 3 operators, we can define multiple Boolean algebras
- Here we focus on two valued Boolean algebra introduced & formalized by C. Shannon (1938) as a follow-up of the work from Edward V. Huntington (1904)
  - Set $B = \{0, 1\}$ is defined with:
    - 0 – `FALSE` and 1 – `TRUE`
  - And two binary operators:
    - $+$ inclusive or (`OR`) and $\cdot$ (`AND`)
  - Attention!!! – In Boolean algebra operators $\cdot$ & $+$ are not arithmetic, but logical operators, so $1 + 1 \neq 2$

[1] Also represented with a horizontal line above the variable symbol, easier for pen & paper.

Boolean algebra: definition of AND/OR operators

- Boolean operators are defined using Truth Tables $\rightarrow$ they indicate operator result for all possible combinations of arguments
- We examine all possible combinations for arguments x & y: table will contain 4 lines (2^2), and each operator has it's own column

x	y	$x \cdot y$	$x + y$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

- Note that we can manufacture electronic device to calculate the above $\rightarrow$ logic gate for which the arguments appear as inputs, and Boolean function evaluation as output

Six axioms of Boolean algebra

- For a quadruplet $\{B, ', \cdot, +\}$, to be a Boolean algebra, following axioms needs to be satisfied:
  - Axiom 1. Closure $\rightarrow$ B is closed for $+$ and for $\cdot$
    - In another words, result of operations is in B
  - Axiom 2. Neutral element $\rightarrow$ B has a neutral element for $+$ (noted '0') and for $\cdot$ (noted '1')
  - Axiom 3. Commutativity $\rightarrow$ is valid for $+$ and $\cdot$
  - Axiom 4. Distributivity $\rightarrow$ $\cdot$ distributes $+$ and $+$ distributes $\cdot$
  - Axiom 5. Complement $\rightarrow$ of x
  - Axiom 6. In B there are 2 elements x, y such that $x \neq y$
- Associativity is the consequence of the previous:

$$\begin{aligned} (a+b)+c &= a+(b+c) \\ (a \cdot b) \cdot c &= a \cdot (b \cdot c) \end{aligned}$$

- It is possible to verify 6 axioms for two-valued Boolean algebra

Basic theorems of Boolean algebra

Th1: Idempotency for $+$ & $\cdot$

$$x + x = x \text{ and } x \cdot x = x$$

Th2: Operations with neutral element

$$x + 1 = 1 \text{ and } x \cdot 0 = 0$$

Th3: Absorption

$$x \cdot (x + y) = x$$

Th4: Involution

$$(x')' = x$$

Th5: Associativity

$$(x + y) + z = x + (y + z) \text{ and } (x \cdot y) \cdot z = x \cdot (y \cdot z)$$

Th6: De Morgan's laws

$$(x + y)' = x' \cdot y' \text{ and } (x \cdot y)' = x' + y'$$

Th7: Consensus

$$x \cdot y + x' \cdot z + y \cdot z = x \cdot y + x' \cdot z$$

Proving propositions in Boolean algebra

- Proving Boolean propositions – formal demonstration of the proposition correction
- Here we use:
  - a) Axioms, theorems – use already proved results to prove new ones
    - We will use this method for exercises as illustration; it is a very good way to integrate different Axioms and Theorems of Boolean algebra (very important!!!); not required as such for the exam
  - b) Truth Tables
    - We can do this since the number of all possible combinations for Boolean arguments is limited and manageable for small number of arguments
  - c) Duality principle
    - Handy way to double the number of proven propositions

a) Proofs using axioms and theorems

- We start with one of the two propositions, so either side of $=$ sign
- We transform the expression using axioms and/or theorems, typically by applying one at the time, and stating exactly which axiom/theorem we are using
- Example – prove Th1. $x+x=x$ using axioms only:

$$
\begin{aligned}
x+x &= (x+x)\cdot 1, && \text{Ax2. Neutral element}\\
&= (x+x)(x+x'), && \text{Ax5. Complement of x}\\
&= x+x\cdot x', && \text{Ax4. Distributivity}\\
&= x+0, && \text{Ax5. Complement}\\
&= x, && \text{Ax2. Neutral element}
\end{aligned}
$$

b) Demonstrations using Truth Tables

- In Two valued Boolean algebra theorems are combinatory problems – we can prove given proposition using Truth Tables by checking the values of both sides of expression for all possible combinations of inputs
- Number of variables determine the number of lines in TT
- For each side of the proposition we will have one TT'; thus there will be 2 TTs for a given proposition
- We calculate the results for both expressions and if they are equal we have proved that the proposition is true

b) Demonstration using TTs – Axiom 1

- Set $B = \{0, 1\}$ is closed: in another words the result of three Boolean operators $+$, $\cdot$, $'$, is in B
- Truth tables are given for the operators:

x	x'
0	1
1	0

x	y	$x \cdot y$	$x + y$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

- The results is indeed in B

b) Demonstration using TTs – Axiom 4

Distributivity:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

x	y	z	$y+z$	$x \cdot (y+z)$	$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0	0	**0**	0	0	**0**
0	0	1	1	**0**	0	0	**0**
0	1	0	1	**0**	0	0	**0**
0	1	1	1	**0**	0	0	**0**
1	0	0	0	**0**	0	0	**0**
1	0	1	1	**1**	0	1	**1**
1	1	0	1	**1**	1	0	**1**
1	1	1	1	**1**	1	1	**1**

b) Demonstration using TTs – De Morgan's law

$(x+y)' = x' \cdot y'$

x	y	$x+y$	$(x+y)'$	x'	y'	$x' \cdot y'$
0	0	0	**1**	1	1	**1**
0	1	1	**0**	1	0	**0**
1	0	1	**0**	0	1	**0**
1	1	1	**0**	0	0	**0**

$(x \cdot y)' = x' + y'$

x	y	$x \cdot y$	$(x \cdot y)'$	x'	y'	$x' + y'$
0	0	0	**1**	1	1	**1**
0	1	0	**1**	1	0	**1**
1	0	0	**1**	0	1	**1**
1	1	1	**0**	0	0	**0**

c) Duality principal

- In Boolean algebra every result has its dual
- If there is a result S that is true in Boolean algebra, the result $S*$, called dual of S, will be true too
- $S*$ can be obtained by systematically permuting: operators $+$ and $\cdot$ and symbols '0' & '1' in B
- What does this really means is the following: duality principal asserts that Boolean algebra is unchanged when all dual pairs are interchanged
- In another words proving S automatically proves $S*$

c) Duality principal – examples

- Example 1
  - Assume the following result S is true:

$$\forall x, x \in B, x + x = x$$

  - Dual of S, $S*$ is true too:

$$\forall x, x \in B, x \cdot x = x$$

- De Morgan Laws
  - The following is true:

$$(x + y)' = x' \cdot y'$$

  - Dual is true too:

$$(x \cdot y)' = x' + y'$$

c) Duality principle generalized to n variables

- It is possible to generalize the previous result on n variables:

$$\forall x_i, x_i \in B, S(x_1, \ldots, x_n) \rightarrow S*(x_1, \ldots, x_n)$$

- Example – Extension of De Morgan laws on n variables

  If:

$$(x_1 + \ldots + x_n)' = x_1' \cdot \ldots \cdot x_n'$$

  Then:

$$(x_1 \cdot \ldots \cdot x_n)' = x_1' + \ldots + x_n'$$